

Remark: From the definition of a valid argument form we see that the argument form with premises $p_1, p_2, \dots, p_n$ and conclusion q is valid exactly when $\boxed{[(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q]}$ is a tautology.

The key to showing that an argument in propositional logic is valid is to show that its argument form is valid.

1.6.3 Rules of Inference for Propositional Logic

- The truth table approach to show that an argument form is valid is tedious. For example, when an argument form involves 10 different propositional variables, to use a truth table to show this argument form is valid requires $2^{10} = 1024$ different rows.
- Instead, we can first establish the validity of some relatively simple argument forms, called rules of inference. These rules of inference can be used to construct more complicated valid argument forms.
- The tautology $\boxed{(p \wedge (p \rightarrow q)) \rightarrow q}$ is the basis of the rule of inference called modus ponens, or the law of detachment. This tautology leads to the following valid argument form.

Example 2: Determine whether the argument given here is valid and determine whether its conclusion is true because of the validity of the argument.
 "If $\sqrt{2} > \frac{3}{2}$, then $(\sqrt{2})^2 > (\frac{3}{2})^2$. We know that $\sqrt{2} > \frac{3}{2}$. Consequently, $(\sqrt{2})^2 = 2 > (\frac{3}{2})^2 = \frac{9}{4}$."

Solution: Let p be " $\sqrt{2} > \frac{3}{2}$ " and q be " $(\sqrt{2})^2 > (\frac{3}{2})^2$ ".
 p is False, since $\sqrt{2} \approx 1.414$ and $\frac{3}{2}$ is 1.5 $\therefore \sqrt{2} > \frac{3}{2}$ is False.
 By defn. of the conditional, $(p \rightarrow q)$ is True. This argument is valid because its modus ponens, i.e. premise: $p \rightarrow q$ and p